The plain additive-quadruple implication on a finite interval

Author proposal: /root/astra_v14_baseline_theta

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Scope. This local proof supplies the exact plain correlation statement of GTZ Proposition 8.4, p. 1277, needed by the Section 13 join. It does not use Proposition 8.3 or infer plain bias from lower-degree bias. The argument is the elementary two-step Cauchy–Schwarz pattern; the finite domains and constants are written explicitly. No source error in Proposition 8.4 is asserted. This author packet requires independent review.

1 Precise finite assertion

Put $J = \{1, \dots, N\}$, $I = \{-N, \dots, N\}$, and suppose $H \subseteq I$ has $|H| \geq \alpha |I|$ for fixed $\alpha > 0$. Let $f : J \to \mathbb{C}$ be 1-bounded and extend it by zero to $\mathbb{Z}$. For each $h \in H$ let $\chi_h : \mathbb{Z} \to \mathbb{C}$ be globally bounded by a fixed $B > 0$, and assume

$$\left| \mathbb{E}_{n \in J} f(n+h)\overline{f(n)}\chi_h(n) \right| \geq \delta > 0.$$

The values outside J are part of the data and are not truncated.

Theorem 1. *Write $a = \alpha\delta/3$ and $\eta = a^4/4$. There are at least*

$$\frac{\eta}{2B^4}(4N+1)(2N+1)^2$$

distinct additive quadruples $(h_1, h_2, h_3, h_4) \in H^4$, $h_1 + h_2 = h_3 + h_4$, for which

$$\left| \mathbb{E}_{n \in J} \chi_{h_1}(n)\chi_{h_2}(n + h_1 - h_4)\overline{\chi_{h_3}(n)\chi_{h_4}(n + h_1 - h_4)} \right| \geq \eta/2.$$

In particular their number is a positive fixed multiple of N^3.

2 Two exact Cauchy–Schwarz steps

For each $h \in H$ choose a scalar c_h of modulus one so the given average multiplied by c_h is real and at least δ. Set $u_h = c_h\chi_h$ on H, and $u_h = 0$ for every $h \notin H$. The functions remain defined on all integers. Then

$$\mathbb{E}_{h \in I}\mathbb{E}_{n \in J} f(n+h)\overline{f(n)}u_h(n) \geq \alpha\delta.$$

Let $M = J + I = \{1 - N, \dots, 2N\}$, so $|M| = 3N$. Set $m = n + h$. The preceding average is exactly

$$3\mathbb{E}_{m \in M} f(m)\mathbb{E}_{h \in I}\overline{f(m-h)}u_h(m-h).$$

Terms with $m - h \notin J$ vanish because f was zero-extended. Cauchy–Schwarz in m, using $|f| \leq 1$, therefore yields

$$\mathbb{E}_{m \in M} \left| \mathbb{E}_{h \in I}\overline{f(m-h)}u_h(m-h) \right|^2 \geq a^2. \tag{1}$$

Expand the square and put $h' = h + k$, $n = m - h$. Let $K = \{-2N, \dots, 2N\}$. The expanded left side equals

$$\frac{N}{|M|}\frac{|K|}{|I|}\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}\overline{f(n)}f(n-k)T(k,n), \qquad T(k,n) = \mathbb{E}_{h\in I}u_h(n)\overline{u_{h+k}(n-k)}.$$

This is an equality with the following domain checks. The factor $\overline{f(n)}$ forces $n \in J$, and then $m = n + h \in M$ automatically for $h \in I$. The condition $h + k \in I$ is supplied by $u_{h+k} = 0$ outside $H \subseteq I$. Thus no additional cutoff in $n-k$ is inserted into T. Its values there are legitimate because the original functions are globally defined. The remaining factor $f(n-k)$ stays in the bounded weight.

The prefactor is positive and at most two. By (1), the absolute value of the displayed average without prefactor is at least $a^2/2$. Cauchy–Schwarz in (k, n), dropping only the bounded weight $\overline{f(n)}f(n-k)$ by an inequality, gives

$$\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}|T(k,n)|^2 \geq a^4/4 = \eta.$$

Expand this square with independent $h, h' \in I$. It is the average over $(k, h, h') \in K \times I^2$ of

$$C(k,h,h') = \mathbb{E}_{n\in J}u_h(n)u_{h'+k}(n-k)\overline{u_{h'}(n)u_{h+k}(n-k)}.$$

The average is real and at least η, while $|C| \leq B^4$. Consequently at least a fraction $\eta/(2B^4)$ of triples have $|C| \geq \eta/2$. Indeed the complementary contribution to the average of $|C|$ is at most $\eta/2$, and the contribution of the selected fraction is at most that fraction times B^4.

Such a triple has $h, h', h + k, h' + k \in H$, since otherwise $C = 0$. Set

$$(h_1, h_2, h_3, h_4) = (h, h' + k, h', h + k).$$

It is an additive quadruple and $h_1 - h_4 = -k$. The map is injective, with inverse $h = h_1$, $h' = h_3$, $k = h_4 - h_1$. The factors c_h contribute only a constant phase of modulus one to C, so they do not change its absolute value. This proves the theorem, including the exact unweighted average on J.

3 Vector families, internal limits and actual consumer

For vector-valued χ_h of fixed finite dimension D, a positive Euclidean correlation implies that some component has magnitude at least $\delta/\sqrt{D}$. Finite pigeonhole retains a component on a subfamily of density at least α/D. Apply the scalar theorem there. The scalar output is a component of the full tensor product, so it implies plain vector bias. A finite list of dimensions can first be padded into one fixed dimension. For a bounded f with a fixed bound other than one, divide it by that bound, changing δ by a fixed factor only.

Apply the finite assertion to representatives of an internal family, with uniform standard bounds, density and correlation. Choices of components and phases can be made for each finite family and then passed to the ultralimit. All constants above are positive standard numbers, so the selected quadruples have positive standard density on the N^3 scale and the correlation is positive by a standard amount. This is exactly the plain Proposition 8.4 output.

In Section 13 the input to this theorem is the entire bounded product that includes the selected lower nilcharacter. Keeping it inside χ_h throughout the calculation preserves that lower information. The theorem requires no degree bound and introduces no lower-degree test factor. It therefore supplies the stronger plain statement that cannot be replaced merely by the degree-biased LP03 output.

The bounded application audit checks the original J, the enlarged $M = J + I$, the parameter range K, the exact inverse reindexing and the global domain of χ_h. A possible artificial cutoff $1_J(n-k)$ is unnecessary: $f(n-k)$ is removed by the second Cauchy–Schwarz inequality, not by claiming an equality after erasing its support. This resolves the domain question without classifying the published Proposition 8.4 as false or defective. The modified Proposition 7.3, its nilsequence-specific factorization and the remaining Section 13 hypotheses are outside this theorem's scope.